

Day 4 Assignment – Vital Sign Analysis

One paragraph on a chosen vital sign, what ranges denote abnormal conditions.

Chosen Vital Sign: Respiratory Rate (RR)

The respiratory rate counts how many breaths a person takes in 60 seconds, measured in breaths/min. It is an essential early warning metric in emergency triage. A healthy adult breathes at a stable baseline of 12 to 20 breaths/min, and deviations are highly abnormal and are signs of physiological distress (Lockett 2022). Tachypnea (>20 breaths/min), shallow quick breaths, often indicates that the patient's respiratory system is actively in distress, struggling to keep up with metabolic demands and supplying oxygen. Rates crossing 24 breaths/min are a call for immediate high-priority triage escalation. This is frequently seen in cases of chronic obstructive pulmonary disease, pneumonia or asthma – illnesses that limit how much air the lungs are taking in (WebMD 2024). Bradypnea (<12 breaths/min), slower breaths than normal, points to a dangerous slowing of the central nervous system and a deprivation of oxygen in the body. This indicates potential respiratory failure and toxic metabolic states, often in cases like carbon monoxide poisoning or drug overdose (WebMD 2024). In the Emergency Department, it is important to track RR deviations to allow the triage system to catch unstable patients whose bodies are working dangerously hard to maintain homeostasis before their other vitals collapse.

References

Lockett, Eleesha. 2022. "What is a Normal Respiratory Rate for Adults and Children?" Healthline.
<https://www.healthline.com/health/normal-respiratory-rate#what-it-measures>

WebMD. 2024. "Types of Breathing Problems, Explained."
<https://www.webmd.com/lung/breathing-problems>